Oregon State University
College of Engineering

ENGR 103: Engineering Computation and Algorithmic Thinking

Homework Assignment #2 – Equipment Fraction Calculation

Note: Part 1 and Part 2 have different due dates (**see page 5**)

ASSIGNMENT OVERVIEW AND OUTCOMES

The focus of this assignment is on solving simple problems using programming language data types, variables, and arithmetic operations. In this assignment, you will:

- Develop a program to read data from the standard input and produce data to the standard output.
- Translate a given mathematical expression into equivalent, syntactically correct programming statements.
- Write a program that conforms to a programming style specified by the instructor.

Be sure to document your program properly. Furthermore, consult the **Python Style Guidelines** document available on Canvas for style expectations. A well-written program is easier to debug, maintain over time, and extend as new requirements arise.

BACKGROUND

When planning a factory for producing ***widgets***[1], machines must be purchased and installed for processing these widgets. Many different types of machines are usually required. The number of each type of machine to be installed in a factory must be determined.

During processing in a factory, it is not uncommon to produce widgets (i.e., output) that do not meet the quality requirements of that step. For example, a cut may be too long or too short, a chemical may have been over-processed or under-processed, etc. When low-quality widgets cannot be fixed, they must be ***scrapped*** (i.e., removed from further processing steps and discarded). When this happens, more widgets must be started than are ultimately required.

Knowing the number of good widgets required to be produced, and the anticipated amount of scrap from each step, the total number of widgets that need to be started can be calculated.

Equipment Fraction Calculations

The number of widgets that need to be started at step 1 of a manufacturing process can be calculated using equation 1:

$$I_1 = \frac{O_n}{\prod_{i=1}^{n}(1 - d_i)} \tag{1}$$

Where:

$I_1 =$ The total number of widgets to start processing.
$O_n =$ The total number of good widgets needed to complete processing.
$d_i =$ The percentage of widgets expected to be scrapped at processing step i for steps 1 to n.

[1] ***Widget*** is a generic term meaning any small device - pick your favorite application: automotive, semiconductors, batteries, medical supplies, etc.

Once the number of widgets to start during step 1 has been determined, the number of machines required for producing widgets at step 1 can be calculated using equation 2:

$$F = \frac{SQ}{EHR} \tag{2}$$

Where:

$F =$ Number of machines required per "shift" (i.e., *fractional* machines required).

$S =$ Standard time per widget (in hours).

$Q = I_1 =$ # of widgets to produce over a fixed time period (i.e., quota of output per shift).

$E =$ Actual machine performance expressed as a percentage of standard time (i.e., the efficiency of the machine). This is a unitless number in the range 0 to 1.

$H =$ Amount of time available per machine over a fixed time period (i.e., hours per shift).

$R =$ Reliability of a machine, expressed as a percentage of uptime. This is a unitless number in the range 0 to 1.

Note that in equation 2, the value for Q will be the value of I_1 calculated previously with equation 1. Also, note that the time period used for Q and H must be the same. For example, it could be a shift, a week, or a month.

<u>Example</u>

A widget manufacturing process has two steps (i.e., 1 and 2). The number of good widgets needed (i.e., the output) after step 2 is 100. There is 2% scrap at step 1, and 3% scrap at step 2. Therefore, $O_2 = 100$, $d_1 = 0.02$, $d_2 = 0.03$. First, use equation 1 to calculate I_1 as:

$$I_1 = Q = \frac{100}{[(1 - 0.02) * (1 - 0.03)]}$$

Finally, use equation 2 to calculate F using the value of Q calculated in the step above.

PROGRAM TO WRITE

Assume a widget manufacturing process that has three steps. Each processing step generates scrap. You must write a computer program that calculates the required number of fractional machines (i.e., F) for step 1 **only**.

You will need input from the user for scrap percentages (i.e., d_1, d_2, d_3) and for the number of good widgets that will be required to come out of step 3 (i.e., O_3). You will also need to get input from the user for all the parameters required in the equipment fraction calculation (i.e., equation 2). Note that Q is going to be calculated by your program based on d_1, d_2, d_3, and O_3.

Next, write instructions that will calculate the fractional number of machines required (i.e., F) using equation 2. Print the value of F back to the user as:

1. A decimal with two digits of precision, and
2. A value rounded up to the next whole number.

Hint: If you need to look up how to print two digits of precision or how to round up, you can search online!

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PART 1 – DESIGN (25 PTS)

First, you will begin your program design by applying Pólya's problem-solving steps.

Step 1 – Understanding the Problem (7.5 pts)

- Do you understand everything in the problem? List anything you do not fully understand.
- What are the functional requirements of the program, i.e., what does it need to do?
- What assumptions are you making?
- What are the inputs, outputs, etc.?

Step 2 – Devise a Plan (10 pts)

- What are the decisions that need to be made in this program?
- What is the sequence of steps you need to complete?
- How are you going to calculate the equipment fraction (i.e., F)?

Based on your answers above, **provide an algorithm as *pseudocode* or as a *flowchart*** showing the specific steps that are needed to create the required computer program. **Be very explicit!!!** No error checking is required in Homework Assignment #2.

Step 4a – Looking Back (7.5 pts)

Create a test plan (on paper) with test cases (i.e., good, bad, and edge cases).

- What are the good, bad, and edge test cases for **ALL** input in the program?
- What do you expect the results to be **in each test case**?

PART 2 – PROGRAM (75 PTS)

After completing Part 1 (i.e., program design), you will write the required computer program in Python. You will also run your Python computer program with different user inputs to ensure it works correctly.

Step 3 – Carry out the plan (30 pts)

Write the computer program specified. Your computer program will be evaluated based on functionality (20 pts) and style (10 pts).

Step 4b – Revisited Looking Back (10 pts)

Revisit your test plan with the test cases below.

- Does the program do the calculations correctly?
- Can you test other cases, and do the results make sense?

Example Test Cases

These are just examples. **You need to create your own test cases**.

Test Case 1:

```
What is the scrap percentage for step 1? Enter it as a number between 0 and 1: 0.1
What is the scrap percentage for step 2? Enter it as a number between 0 and 1: 0.05
What is the scrap percentage for step 3? Enter it as a number between 0 and 1: 0.02
What is the number of good widgets required as output for step 3 per shift?: 1000
What is the standard time per part (in hours)?: 2.3
What is the number of hours per shift (in hours)?: 8
What is the machine reliability percentage? Enter it as a number between 0 and 1: 0.99
What is the machine efficiency percentage? Enter it as a number between 0 and 1: 0.88

The calculated required fractional machines is 393.85 machines.
The calculated whole number of machines is 394 machines.
The calculated whole number of machines is 394.0 machines.
```

Test Case 2:

```
What is the scrap percentage for step 1? Enter it as a number between 0 and 1: 0.01
What is the scrap percentage for step 2? Enter it as a number between 0 and 1: 0.04
What is the scrap percentage for step 3? Enter it as a number between 0 and 1: 0.003
What is the number of good widgets required as output for step 3 per shift?: 9000
What is the standard time per part (in hours)?: 0.87
What is the number of hours per shift (in hours)?: 12
What is the machine reliability percentage? Enter it as a number between 0 and 1: 0.95
What is the machine efficiency percentage? Enter it as a number between 0 and 1: 0.99

The calculated required fractional machines is 732.18 machines.
The calculated whole number of machines is 733 machines.
The calculated whole number of machines is 733.0 machines.
```

Demo Program to TA (35 pts)

You need to schedule time with a TA to conduct a “live” demo of your computer program.

SUBMISSION AND GRADING

Part 1 – Design (25 pts)
Due on Monday, January 19, by 11:59 pm

You will submit a PDF file to Gradescope with answers to the three steps in Part 1 (i.e., the design – see page 3). As part of the upload/submission process, you **must** select the pages in the PDF file corresponding to each specific step of the design process. See the instructor or the TAs if you have any questions.

Part 2 – Computer Program (in Python) and Revisited Looking Back (40 pts)
Due Monday, January 26, by 11:59 pm

Part 2 has two deliverables. As the first deliverable, you will submit **only** the Python code file (*.py) to Gradescope. As the second deliverable, you will submit a PDF file that shows (1) the output produced by your computer program for the test cases, and (2) your answers to the questions in Step 4b (i.e., Revisited Looking Back – see page 4).

Second and Third Weeks – Demo Program to a GTA/ULA (35 pts)

During the weeks of **January 26** and **February 2**, you **must** sign up for a 10-minute demo slot with a TA **using the Canvas calendar**. You are encouraged to sign up for your demo slot as soon as the homework assignment is released.

A TA will run the code you submitted via Gradescope on their computers with the test cases you provided and with additional test cases. Also, they will ask you questions about your code and methods to ensure that you understand their function.

Failing to show up on time for a scheduled demo will result in a 15-point deduction, and another demo must be scheduled. Failing to demo the program at all will result in all 35 points being lost.